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Exploring Simulation Methods to Counter Cyber-Attacks on the Steering Systems of the Maritime Autonomous Surface Ship (MASS)

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Abstract

This paper presents a simulation-based investigation into control strategies for mitigating the consequences of cyber-assault on the steering systems of the Maritime Autonomous Surface Ships (MASS). The study focuses on two simulation experiments conducted within the Simulink/MATLAB environment, utilizing the catamaran "Nymo" MASS mathematical model to represent vessel dynamics. Cyber-attacks are modeled as external disturbances affecting the rudder control signal, emulating realistic interference scenarios. To assess control resilience, two configurations are compared during a representative turning maneuver to a specified heading: (1) a Proportional–Integral–Derivative (PID) regulator augmented with a Least Mean Squares (LMS) adaptive filter, and (2) a Nonlinear Autoregressive Moving Average with Exogenous Input (NARMA-L2) neural network regulator. The PID and LMS configurations aim to enhance the disturbance rejection capabilities of the classical controller through adaptive filtering, while the NARMA-L2 approach represents a data-driven, nonlinear control alternative. Simulation results indicate that although the PID and LMS setups demonstrate improved performance over standalone PID in the presence of cyber-induced disturbances, the NARMA-L2 controller exhibits superior adaptability, accuracy, and robustness under adversarial conditions. These findings suggest that neural network-based control offers a promising pathway for developing cyber-resilient steering systems in autonomous maritime vessels.

Keywords: catamaran; MASS; cyber-attack; nonlinear system; PID controller; LMS filter; NARMA-L2 controller

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1. Introduction

The marine industry is going through a substantial shift, owing to the widespread adoption of smart shipping technologies that combine connectivity, robotics, large-scale data analytics, and computational intelligence [1]. These innovations aim to enhance competitiveness, operational safety, and environmental sustainability. Modern ships are increasingly equipped with a variety of sensors, onboard data processors, and data-driven decision-making systems based on deep learning algorithms. Thanks to high-bandwidth connectivity, vessels can now be remotely monitored and operated, laying the groundwork for greater autonomy at sea. However, further technical breakthroughs have made it

possible for ships to function independently. A vessel can maneuver autonomously or request human intervention depending on a real-time interpretation of sensor data [2].

The emergence of MASSs represents a pivotal shift in maritime operations. MASSs are digital–physical structures enabled by the convergence of novel real-time Information and Operational Technologies (IT/OT).

The OT encompasses a broad range of programmable controllers and embedded systems that manage physical processes on board, while the IT handles data management and administrative functions. Both domains are critical to MASS operation, and both are increasingly susceptible to cyber-attacks [3].

MASSs have several advantages: they may be constructed to be cleaner on fuel, less heavy, more affordable to operate, wind-resistant, and aerodynamically efficient. Nevertheless, the path to widespread MASS adoption does not lack impediments. Legal and regulatory frameworks pose significant limitations. For example, the Convention on the International Regulations for Preventing Collisions at Sea (COLREG) primarily governs crewed vessels, making rules such as "proper look-out" (Rule 5) and "good seamanship" (Rule 8) difficult to interpret or apply in the context of unmanned systems. Furthermore, the International Ship and Port Facility Security (ISPS) code requires the inclusion of designated security personnel on the vessel (A/2.1.6), thereby banning the deployment of entirely crewless ships [4].

In addition to regulatory issues, MASSs face a range of practical and technical concerns. Human factors such as programming and coding errors or incorrect responses to rare or unforeseen scenarios can limit the reliability of autonomous systems. Interactions between manned and unmanned vessels on busy waterways pose operational dangers, as does a failure to identify smaller or low-visibility vessels. Environmental variables that involve storms, high waves, high levels of tides, and dense glaciers all pose challenges to autonomous navigation systems. Hardware failures, including propulsion, rudder, or sensor malfunctions, as well as software or communication breakdowns, can significantly compromise the vessel's safety and autonomy [5–7].

Cybersecurity is perhaps one of the most critical and complex issues facing MASS development. Cyber problems, either intentional (integrity breaches) or unintentional (operational failures), endanger human lives, marine infrastructure, and the environment. Common cyber threats include attacks on onboard IT and OT systems, spoofing or jamming of Global Navigation Satellite System (GNSS) signals, manipulation of (Satellite-) Automatic Identification System ((S-)AIS) data, and the disruption of connectivity with shore-based operation centers [8]. Alarmingly, most marine insurance policies do not currently cover the consequences of such cyber incidents [9].

Given the increasing frequency and sophistication of cyber-attacks, it is essential to implement layered and proactive cybersecurity strategies. These include regular system audits, use of firewalls and anti-virus software, encryption of sensitive communications, maintaining up-to-date software versions, and developing dynamic protection mechanisms that can evolve alongside new threat vectors. More importantly, ensuring the cyber-resilience of MASSs will require a combination of technical innovation, regulatory adaptation, and operational vigilance [10].

This study investigates how cyber-attacks can compromise the steering control systems of Maritime Autonomous Surface Ships (MASSs) and evaluates the effectiveness of different control strategies in mitigating these threats. More precisely, the following research questions are addressed:

(1) How do cyber-attacks, modeled as signal disturbances, affect the trajectory and stability of steering systems in the MASS?

(2) To what extent can classical control methods (e.g., PID with LMS filtering) mitigate such cyber threats in simulated environments?
(3) How effective is a machine-learning-based NARMA-L2 controller in maintaining navigation stability under adversarial cyber conditions compared to classical methods?

To address this research issue, the paper begins with an overview of the topic, followed by a literature review, mathematical modeling of the Nymo MASS, and simulation experiments designed to neutralize cyber-attacks on its steering control system. The remaining part of the paper contains a comparison of a classical PID controller, a PID controller enhanced with LMS adaptive filters, and a NARMA-L2 neural network controller, demonstrating that the NARMA-L2 approach delivers superior performance under cyber-threat conditions. The article concludes with key findings and outlines directions for further research in enhancing the cyber-resilience and control capabilities of autonomous maritime systems.

2. Literature Review

The development of unmanned and autonomous naval vehicles marks an important change in marine transportation, research, and defense. The development of Autonomous Underwater Vehicles (AUVs) began in the 1950s, evolving from heavy and inefficient early prototypes to sophisticated platforms capable of navigating depths up to 6000 m with advanced obstacle detection and mapping capabilities [11]. Similarly, MASSs have progressed from experimental vessels to commercially operational platforms, including cargo ships, research vessels, and military craft [12]. However, despite these advancements, fully autonomous underwater and surface vessels capable of deep and hazardous missions remain a work in progress, with numerous technical, legal, and organizational challenges still to be resolved [13].

International regulatory authorities, like the International Maritime Organization (IMO), are actively working on the legal and safety frameworks necessary for the safe integration of MASSs into existing marine traffic. The IMO anticipates finalizing non-mandatory MASS regulations before 2025, which will lay preparatory measures for a comprehensive mandatory regulatory framework expected to be implemented post-2030 [14]. Classification societies like Loyd's Register have established autonomy levels ranging from manual operations (AL0) to fully autonomous vessels without human intervention (AL6), providing a standardized taxonomy for assessing unmanned marine systems [15]. Current examples of operational MASS include the Yara Birkeland autonomous cargo vessel, SEA-KIT's USV Maxlimer, the US Navy's Large Unmanned Surface Vehicles (LUSVs) and Extra Large Unmanned Undersea Vehicles (XLUUVs), the Chinese autonomous mother ship Zhu Hai Yun, the Mayflower autonomous research vessel, and ongoing projects such as South Korea's KASS and Rolls-Royce's AAWA, aiming for full autonomy by 2035 [16].

While autonomous vessels offer significant operational advantages, their reliance on digital navigation and control systems makes them vulnerable to cyber-attacks, which pose critical safety and operational risks. The most frequent targets in maritime navigation systems are the Global Positioning System (GPS) and (S)-AIS, both susceptible to spoofing, jamming, and data manipulation attacks [17]. Spoofing attacks can create false location information, misleading vessel navigation systems, and potentially cause collisions or grounding. Jamming disrupts signal reception, rendering GPS receivers inoperative unless countermeasures are employed, such as filtering using computational intelligence [18].

Aiming to enhance resilience, hybrid GNSS receivers combining multiple satellite constellations (GPS, GLONASS, BeiDou, Galileo) and augmentation systems such as the European Geostationary Navigation Overlay System (EGNOS) are employed, alongside inertial navigation systems and onboard mathematical models of vessel dynamics and

environmental influences [19]. These alternative approaches provide redundancy, allowing autonomous vessels to maintain navigation capabilities despite compromised satellite data. However, these solutions can be costly and complex, particularly for commercial operations.

Beyond navigation systems, other maritime cyber vulnerabilities exist in Supervisory Control and Data Acquisition (SCADA) systems, communication protocols, and monitoring architectures. The (S-)AIS, used for ship identification and collision avoidance, is prone to spoofing and fraudulent message injection, which can lead to erroneous routing or misrepresentation of vessel types and locations [20,21]. To counteract these threats, layered defenses incorporating radar, advanced camera systems, and verification through updated electronic charts are necessary.

Cyber-resilience, defined as the ability of a system to maintain critical functions despite cyber-attacks, is becoming an essential characteristic for MASSs operating within the highly interconnected and safety-critical maritime ecosystem [22]. Notwithstanding, the shipping industry's traditional nature and hesitancy towards rapid adoption of new technologies complicate the timely implementation of robust cybersecurity measures. Research on maritime cybersecurity has expanded significantly since 2011, focusing on risk assessment, safety engineering, and systemic approaches such as System Theoretic Process Analysis (STPA), which enables holistic evaluation of MASS risks [23,24]. Probabilistic modeling techniques, including Bayesian networks, have been applied to analyze vulnerabilities and guide mitigation strategies [25]. The Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege (STRIDE) threat model is widely utilized to classify cyber threats specifically impacting autonomous ship systems and communication networks [26]. These models identify denial of service and spoofing as important hazards to navigation, identification, and control operations. Within this framework, we must include the Zero Trust Architecture paradigm alongside the MITRE ATT&CK scheme, for which the following applies: "don't trust, constantly check" [27,28]. This is an essential part of a contemporary security plan that helps various sectors to actively evaluate the evolving landscape of cyberthreats and guarantee a more secure and resilient digital future. Furthermore, ref. [29] provides a thorough examination of cyber resources, hazards, risk evaluation, and mitigation strategies for autonomous vessels. This article refers to the most recent cyber-security standards published by the IMO and the Baltic and International Maritime Council (BIMCO), which served as the foundation for this investigation. It also comprises a cyber-risk assessment based on Hazard Identification (HAZID) that was carried out in accordance with smart ships.

Advances in cybersecurity research emphasize the integration of Machine Learning (ML) and Artificial Intelligence (AI) for early breach identification, anomaly recognition, and adaptive defense in maritime autonomous systems [30]. For example, frameworks employing deep autoencoders and clustering algorithms are developed to discover cyber irregularity in message streams [31]. Furthermore, study [32] introduces event-triggered mechanisms and robust control strategies to ensure synchronization despite cyber threats on communication channels in master–slave neural networks. Similarly, paper [33] discusses how cyber attackers can exploit open communication networks to disrupt networked control systems by intentionally blocking data packets, causing malicious packet dropouts. Despite these developments, there is a recognized shortage of maritime-specific datasets and realistic simulation environments necessary to train and validate such ML/AI-based systems effectively [34,35].

Complementary to technical defenses, broader supply chain security considerations highlight the prevalence of Distributed Denial of Service (DDoS) attacks and malware targeting maritime infrastructure, urging the implementation of Industry 4.0 assets to enhance cyber defenses [36]. Ports, shipping companies, and maritime supply chains

have become frequent targets of ransomware and data exfiltration attacks, underscoring the importance of comprehensive cybersecurity strategies extending beyond individual vessels [37,38]. Special attention has also been directed towards the vulnerabilities of SCADA systems, which are integral to maritime operations. Their real-time requirements and critical role in operational safety demand tailored cybersecurity solutions, including anomaly detection and rapid response capabilities [39,40].

Simulation studies on autonomous vessel controllers have demonstrated that cyber intrusions targeting rudder and thruster control systems can cause severe disruptions in vessel maneuverability and safety. Kalman filters combined with PID controllers have been proposed and validated through Simulink/MATLAB experiments and real-vessel case studies as effective mitigation techniques to maintain control under cyber-attack conditions [41–43]. These approaches highlight the necessity of embedding cybersecurity resilience directly into control system architectures. The growing body of research into MASS cybersecurity reveals a multifaceted landscape, combining regulatory efforts, technical innovation, and systemic risk assessment methodologies. The integration of ML/AI, comprehensive threat modeling, and simulation-based testing forms the foundation for advancing autonomous maritime operations' safety and reliability. Continuous collaboration among scholars, the business world, and government entities will be required to develop codified frameworks, datasets, and defense mechanisms to address the changing cyber threat environment confronting autonomous maritime systems.

While prior research has explored maritime cybersecurity broadly, including GNSS spoofing, AIS manipulation, and SCADA vulnerabilities, limited attention has been given to the control-level impacts of cyber intrusions specifically targeting the steering systems of MASS. Existing studies tend to focus on general network or navigation vulnerabilities rather than control surface manipulation. Furthermore, although machine learning and adaptive filtering have been proposed in industrial CPS applications, their role in maritime steering resilience remains underexplored. This study narrows this gap by applying simulation-based attacks directly to the rudder control signal and benchmarking classical PID and LMS filtering techniques against a neural-based NARMA-L2 controller in a dynamic marine setting.

3. Methodology

The simulation experiment in this study involves a single MASS, specifically, the "Nymo" catamaran (Figure 1), modeled using nonlinear dynamics and differential thrust mechanics. The study investigates the cyber-resilience of the vessel's steering control system through a structured methodology encompassing mathematical modeling, realistic cyber-attack emulation, and comparative controller performance analysis. The developed model accurately reflects the ship's maneuvering behavior and serves as a foundation for simulating navigation under both nominal and adversarial conditions.

A dedicated simulation environment was constructed in Simulink/MATLAB to replicate cyber-attack scenarios targeting the rudder control system. These attacks are modeled as Gaussian white noise disturbances superimposed on control inputs, simulating unauthorized interference that may disrupt maneuverability and heading control.

Three control strategies were tested under identical conditions:

- A classical PID controller (serving as the baseline);
- A PID controller augmented with LMS adaptive filters, which improves adaptability and disturbance rejection;
- A NARMA-L2 neural network controller, offering advanced adaptive capabilities.

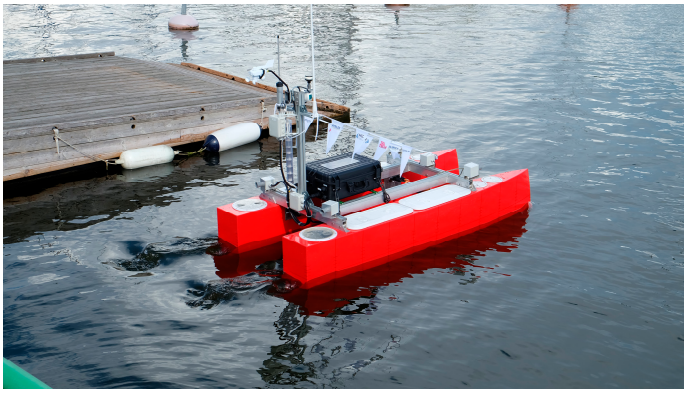


Figure 1. Catamaran-based "Nymo" MASS (Photo by: Tallinn University of Technology).

The experimental scenario involves a turning maneuver to a fixed heading angle (e.g., 45° North–East–Down (NED)), where each control strategy is evaluated based on its ability to maintain trajectory accuracy and system stability in the presence of simulated cyber threats. Performance is assessed using metrics such as heading error, response smoothness, and deviation from the intended course. The results demonstrate that while the PID-based controllers show degraded performance under attack, especially without LMS filtering, the NARMA-L2 controller maintains a smooth and accurate course, exhibiting strong resilience. This confirms its suitability for enhancing the cyber-robustness of autonomous steering systems.

The experiment is a single-vessel, scenario-specific simulation designed to realistically test control system behavior under cyber-assault, contributing to a practical framework for evaluating and improving the security of autonomous maritime navigation.

3.1. Model of a Catamaran MAS

Now, allow us to explore the replica of the catamaran "Nymo" MASS (Figure 2). This catamaran watercraft has two distinct hulls with separate propellers, where F_l and F_r are the port side (left) and starboard side (right) powers, which are produced by differential thrusts, and d is the lateral hull separation (from hull centerline to MASS centerline). Figure 2 depicts the structure of the variable definitions, with the heading angle ψ denoting the orientation of the craft's construction-fixed structure reference to the NED quadrant. For convenience, the current vessel's heading angle φ is taken in a counterclockwise manner from the horizontal plane, i.e., $\varphi = \pi/2 - \psi$. Figure 2 shows that the catamaran's maneuvering is dependent on differential thrust. If F_l equals F_r, the vessel will travel in a straight direction. If F_l and F_r are different, the variation in thrust between the two stern engines creates a turning point, causing the catamaran to change its course.

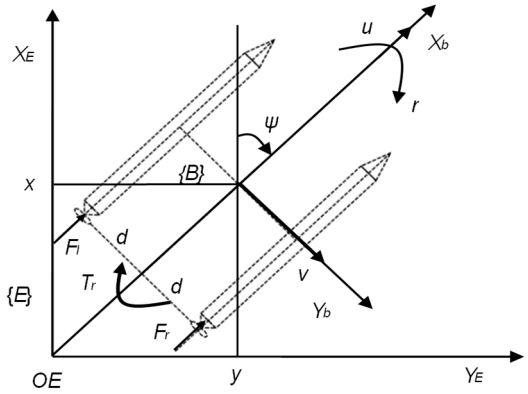


Figure 2. Planar model of catamaran MASS.

The goal is to record the vessel's dynamics and retain the ability to manage its path, ensuring that the catamaran goes to the intended target as quickly and safely as feasible. The kinematics of the model can be described as [44]

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\varphi} \end{bmatrix} = \begin{bmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix}, \tag{1}$$

The coordinates of the catamaran's center of mass in the earth-fixed system are (x,y), φ is the vessel's heading angle, and u, v, r are the surge, sway, and yaw speeds, correspondingly. In the current study, we assume that harming environmental influences such as wind, currents, and waves can be omitted. The system dynamics of the considered simplified mechanical model of the catamaran can be described using the formulas that follow [45]

$$\begin{aligned} m\dot{u} + m_{11}\dot{u} - mvr &= F_x \\ m\dot{v} + m_{22}\dot{v} + mur &= F_y \\ I_{zz}\dot{r} + m_{66}\dot{r} &= N \end{aligned} \tag{2}$$

where m is the mass of the catamaran; m_{11}, m_{22}, m_{66} are the hull-added masses; I_{zz} is the moment of inertia of the catamaran about the Z axis; F_x, F_y, N are the extrinsic power in X direction, extrinsic power in Y direction, and extrinsic moment, respectively. Note that surge, sway, and yaw speeds and accelerations describe a vessel's motion along three axes: surge (forward/backward), sway (side-to-side), and yaw (rotation around the vertical axis). These motions are characterized by both their speed (how fast they are changing position or orientation) and acceleration (how quickly the speed is changing). Therefore, the dots above the variables in (2) denote their derivatives. The extrinsic powers and moment in (2) produced by the propellers can be determined as follows [46]

$$\begin{aligned} F_x &= (1 - t_P)(F_r + F_l) \\ F_y &= 0 \\ N &= (1 + d_{NP})(F_r - F_l)d \end{aligned} \tag{3}$$

where t_P is the thrust deduction factor generated by the propeller in the X direction, d_{NP} is the propeller influence factor in the Y and N directions. Let us observe that in (2), no propeller powers are produced in the Y direction. The propeller's thrusts in (3) are determined as [47]

$$\begin{aligned} F_l &= k_{tl}\left(J_{pl}\right)\rho n_l^2 d_{pl}^4 \\ F_r &= k_{tr}\left(J_{pr}\right)\rho n_r^2 d_{pr}^4 \end{aligned} \tag{4}$$

where J_{pl} and J_{pr} are the propeller advance speed parameters of the left and right propellers; ρ is the density of the seawater; n_l and n_r are the numbers of revolutions of the left and right propellers; d_{pl} and d_{pr} are the diameters of the left and right propellers; and k_{tl} and k_{tr} are the propeller thrust coefficient functions of the left and right propellers, respectively. The integration then yields the heading angle and center of mass coordinates

$$x(\tau) = \int_0^\tau \dot{x}(t)dt, y(\tau) = \int_0^\tau \dot{y}(t)dt, \varphi(\tau) = \int_0^\tau \dot{\varphi}(t)dt, \tag{5}$$

where x(0) = 0, y(0) = 0, and $\varphi(0) = 0$ are the beginning states. The Equations (1)–(5) demonstrate that the heading angle and location coordinates of the MASS $(x, y, \varphi)^T$ can be determined with the specified set of formulas, the intended destination angle, and initial conditions.

3.2. Parameters of the Catamaran MAS

This section assesses the catamaran parameters for MASS "Nymo". Direct measurements can be used to determine this vehicle's subsequent parameters [48].

$$\begin{gathered} m = 225kg, d = 0.405\ m, d_p = 0.18\ m, \\ l_x = 2.56\ m, l_y = 1.08\ m, l_d = 0.36\ m \end{gathered} \tag{6}$$

where catamaran length, width, and draft are denoted by the letters l_x, l_y, and l_d, respectively. It is possible to calculate the magnitudes of t_P and d_{NP} from (3) [46]

$$t_P = 0.25, d_{NP} = 0.25$$

The following formula is used to obtain the parallelepiped's moment of inertia I_{zz} relative to the Z axis [49]

$$I_{zz} = \frac{m}{12}\left(l_x^2 + l_y^2\right) \tag{7}$$

Hence, from (6) and (7), we find

$$I_{zz} = 144.7500\ kgm^2 \tag{8}$$

The parameters of (6) and (8) are linked to the hull-added masses from (2), so [45]

$$\begin{gathered} m_{11} = \frac{l_d m}{2l_x}, \\ m_{22} = \frac{2l_d m}{l_y}\left(1 - \frac{l_y}{2l_x}\right) \\ m_{66} = \frac{2l_d I_{zz}}{l_y}\left(1 - \frac{1.6l_y}{l_x}\right) \end{gathered} \tag{9}$$

Thus, the following values are assigned to the parameters in (9)

$$\begin{gathered} m_{11} = 15.8203\ kg, m_{22} = 118.3594\ kg, \\ m_{66} = 31.3625\ kgm^2 \end{gathered}$$

It is well known that parameter ρ in (4) has the following value:

$$\rho = 1000\ \frac{kg}{m^3}$$

Now, let us assess one of the propellers' F values using Equation (4). For the chosen propeller, the advanced speed coefficient J_p from (4) can be written as [50]

$$J_p = \frac{u}{nd_p} \tag{10}$$

here, u is the surge speed, n is the propeller's number of revolutions, and d_p is the propeller's diameter. A quadratic polynomial of J_p will be used to approximate the propeller thrust coefficient function k_t for simulation purposes,

$$k_t = a_1 J_p^2 + a_2 \tag{11}$$

where the constant polynomial coefficients are denoted by a_1 and a_2. The following system of linear equations provides the polynomial coefficients in (11).

$$\begin{aligned} a_1 J_{p1}^2 + a_2 &= k_{t1} \\ a_1 J_{p2}^2 + a_2 &= k_{t2} \end{aligned} \tag{12}$$

A quadratic polynomial can be used to approximate the propeller thrust coefficient function (k_t) based on the advance ratio (J_p). This approach allows for a simplified representation of the propeller's thrust characteristics, which is particularly useful in simulations and calculations where complex models are not required [51]. Further, it is possible to express this set of Formulas (12) in matrix form so

$$WA_t = K_t \tag{13}$$

where

$$W = \begin{bmatrix} J_{p1}^2 & 1 \\ J_{p2}^2 & 1 \end{bmatrix}, A_t = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, K_t = \begin{bmatrix} k_{t1} \\ k_{t2} \end{bmatrix}$$

The mathematical formula for the vector of polynomial coefficients from (13) is as follows

$$A_t = W^{-1} K_t$$

Using the next experimental data [49]

$$\begin{gathered} u_1 = 0.4450 \ \tfrac{m}{s}, u_2 = 0.8000 \ \tfrac{m}{s}, \\ n_1 = 400 \text{ rpm}, n_2 = 800 \text{ rpm}, \\ k_{t1} = 0.4104, k_{t2} = 0.3954 \end{gathered}$$

we found that

$$A_t = \begin{bmatrix} 0.5699 \\ 0.3321 \end{bmatrix}.$$

Combining (4), (10), and (11), we obtain that the value of propeller thrust can be estimated so that

$$F = \rho d_p^2 \left(a_1 u^2 + a_2 d_p^2 n^2 \right). \tag{14}$$

The number of revolutions can then be computed based on (14), thus

$$n = \sqrt{\frac{F - a_1 \rho d_p^2 u^2}{a_2 \rho d_p^4}}.$$

3.3. PID Control of Course of the MASS

A model of a catamaran is a nonlinear model, and it is challenging to design a simple regulator that, in every situation, adheres to the outlined boundaries. It is possible to consider the thrust of the left propeller, F_l, as a fixed constant. Allowing F_l and F_r to vary would make the system a multiple-input, multiple-output system. This would render the application of PID and NARMA-L2 control strategies inappropriate. Hence, we have

$$F_l = \text{const.} \tag{15}$$

With a choice of (15), the right propeller Fr's thrust is used as a control input to regulate the heading angle φ, transforming the complex control problem into a control problem.

The following structure (see Figure 3) is the result of designing the control system setup to utilize a PID controller to regulate the input variable F_r. PID controllers come in a variety of formats; one potential compensator formula implementation is shown as

$$\mathrm{PID(s)} = \mathrm{P(br - y)} + \mathrm{I}\left(\frac{1}{\mathrm{s}}\right)(\mathrm{r - y}) + \mathrm{D}\left(\frac{\mathrm{N}}{1 + \mathrm{N}\left(\frac{1}{\mathrm{s}}\right)}\right)(\mathrm{cr - y}) \tag{16}$$

where P, I, D, N, b, and c are the parameters of the proportional, integral, derivative, filter coefficient, proportional setpoint weight, and derivative setpoint weight components of the PID controller, respectively; r is the targeted reference; and y is the measured output.

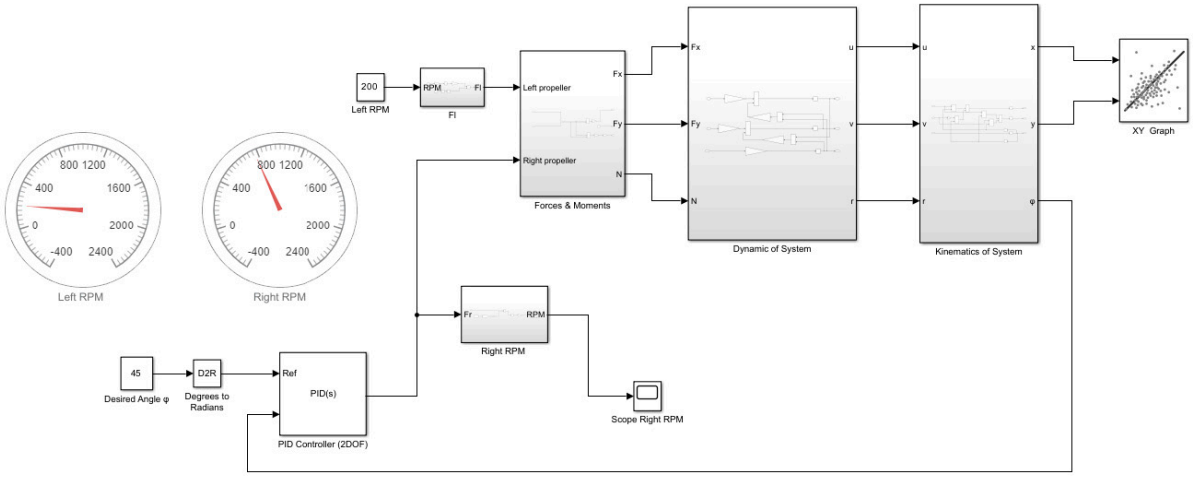


Figure 3. MASS block diagram with PID regulator without irregularities.

Increasing the proportional gain (P) has the effect of proportionally increasing the control signal for the same level of error. The fact that the controller will "push" harder for a given level of error tends to cause the closed-loop system to react more quickly, but also to overshoot more. Another effect of increasing P is that it tends to reduce, but not eliminate, the steady-state error.

The integral term to the controller (I) tends to help reduce steady-state error. If there is a persistent, steady error, the integrator builds and builds, thereby increasing the control signal and driving the error down. A drawback of the integral term, however, is that it can make the system more sluggish (and oscillatory) since, when the error signal changes sign, it may take a while for the integrator to "unwind."

The derivative element to the controller (D) adds the ability of the controller to "anticipate" error. With simple proportional control, if D is fixed, the only way that the control will increase is if the error increases. With derivative control, the control signal can become large if the error begins sloping upward, even while the magnitude of the error is still relatively small. This anticipation tends to add damping to the system, thereby decreasing overshoot. The addition of a derivative term, however, has no effect on the steady-state error.

The general effects of each controller parameter (P,I,D) on a closed-loop system are summarized in Table 1.

Using Simulink/Matlab software (ver. R2024b), the fixed P, I, D, N, b, and c parameters from (16), which are utilized to adjust the controller to a desired behavior, are derived as

$$\mathrm{P} = 5.8134, \mathrm{I} = 0.2722, \mathrm{D} = 30.4924, \mathrm{N} = 5.4540, \mathrm{b} = 0.9213, \mathrm{c} = 0.7141.$$

Table 1. The general influences of the P, I, and D parameters on a closed-loop system.

Type	Rise Time	Overshoot	Settling Time	Sum of Squares Error
P	Decrease	Increase	Small Change	Decrease
I	Decrease	Increase	Increase	Decrease
D	Small Change	Decrease	Decrease	No Change

The path is regulated by the planned control system as it approaches motion with a constant intended heading angle for a constant number of left propeller rotations (n_l = 200 rpm). Figures 4 and 5 display the computer modeling results for the suggested block structure (see Figure 3) for the desired heading angle φ_d = 45 deg. Figure 4 explains the simulation task input signals as the number of right propeller n_r revolutions. Figure 5 displays the resulting path of the specified MASS. In this case, the trend line with the determination coefficient R^2 = 1 is the best approximation of the trajectory by a line with points ideally lying on a straight line.

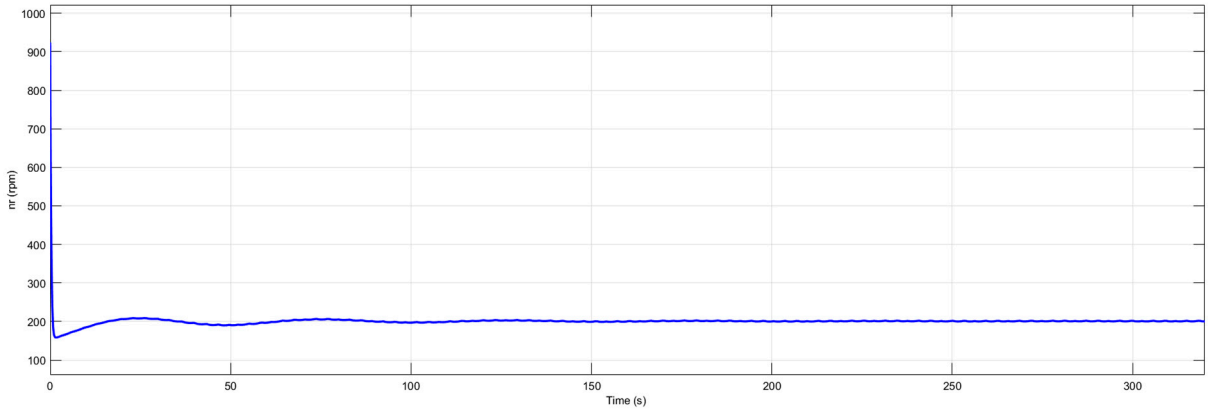


Figure 4. MASS input control signal without irregularities.

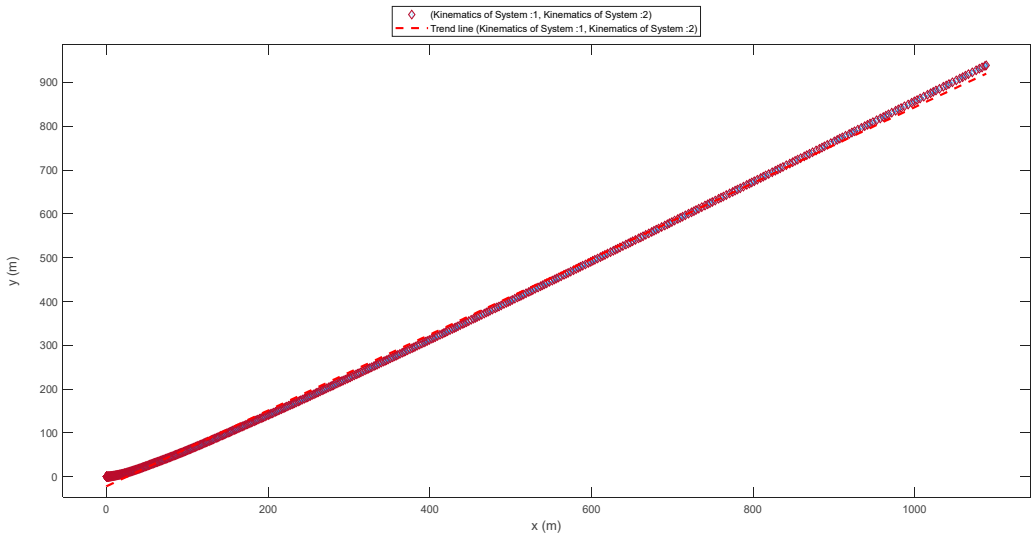


Figure 5. MASS path without irregularities.

3.4. Simulation of Cyberattack Using PID Controller

Cyber–physical systems (CPSs) are digital and mechanical devices that use sensors and actuators to detect and change their state, respectively. The CPSs rely on the accuracy and precision of sensors and actuators to reach a desired state. All parts of CPSs are prone to malfunctions or errors. Also, CPSs are vulnerable to deliberate attacks on both their

cyber and physical components. The errors due to malfunctions or deliverable attacks on CPSs can cause the system to undergo an undesired or possibly dangerous response. The adversary can attack the system at any one of the components of the control system. This attack is propagated through the control system and modifies the value of the control signal in some random manner. This malevolent effect can be modeled as overwriting the original control signal with some random value [52,53]. For instance, the desired rudder command can be overwritten with Gaussian white noise to model the effects of attack on any control system component. The assault spreads across the control system and randomly alters the control signal's value. One way to model the impact of a cyberattack is to inject Gaussian white noise into any part of the control system. The intended output and the actual measurement, φ_a, are connected. Figure 6 shows that φ_d with v is an imposed noise.

$$\varphi_a = \varphi_d(\tau) + \nu(\tau). \tag{17}$$

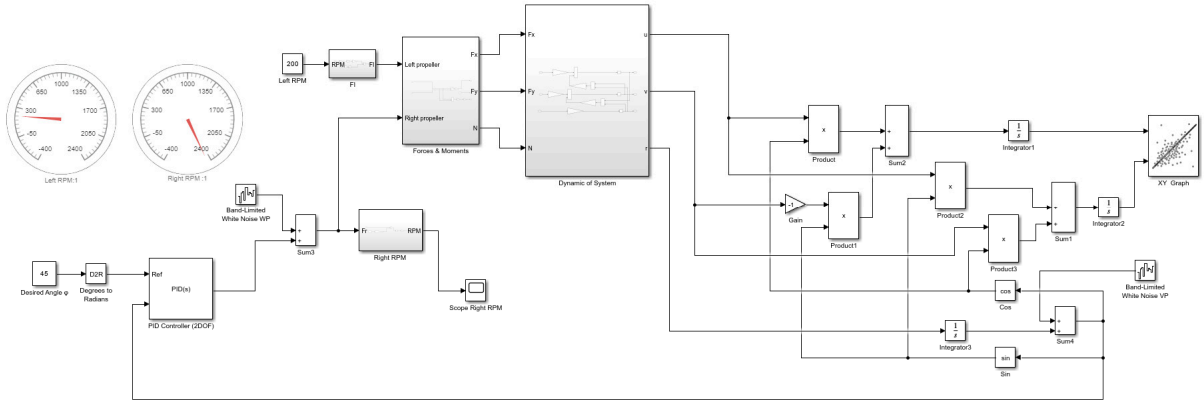


Figure 6. MASS block diagram with PID regulator with irregularities.

Now, examine a system where an interference takes place at the system's input side, as shown in Figure 6.

$$F_{ra}(\tau) = F_{rd}(\tau) + w(\tau) \tag{18}$$

It is challenging to construct a straightforward controller that adheres to a reference in all circumstances since the MASS model is noisy. As previously discussed, the control system configuration uses a PID controller (Figure 6). The white noise powers, $P_W = 1 \times 10^{-7}$ from (18) and $P_V = 5 \times 10^{-5}$ from (17), for the system were intruded into the maneuver. The control signal shadowed by noise, along with the trajectory, is presented in Figures 7 and 8. In this case, the trend line with the determination coefficient $R^2 = 0.9$ is the approximation of the trajectory by a best-fit line which has an angle of 24 deg. Higher rpms (revolutions per minute) (Figure 7) do not necessarily damage the MASS, but consistently running at a high rpm can cause the engine to wear over time. Short bursts of high rpm are generally safe and can even help clean out carbon deposits, but prolonged exposure to high rpms can increase heat and strain engine components. The MASS under such conditions performs chaotic sideways movement off its course near the trend line, which differs significantly from the desired one, and becomes unpredictable and dangerous for other vessels.

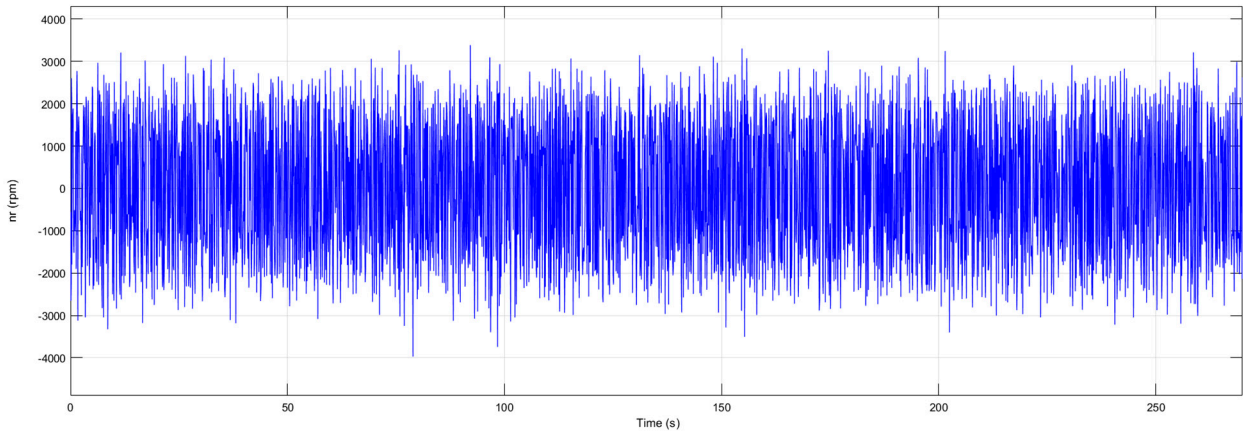


Figure 7. MASS input control signal with irregularities.

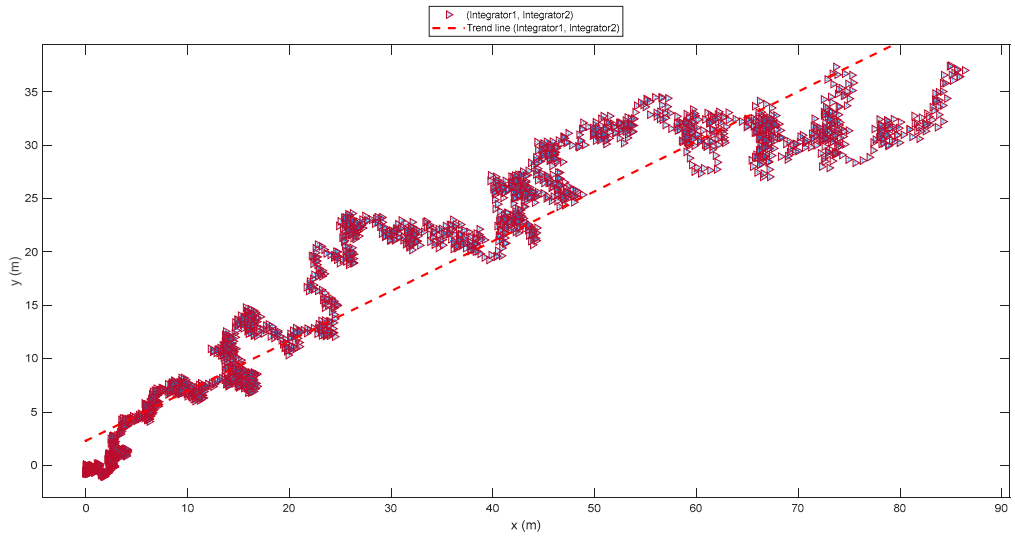


Figure 8. MASS path with irregularities.

3.5. Simulation of Cyberattack Preventing by Using PID Controller and LMS Filters

In this setting, the LMS filters are added to the system's input and output, maintaining the PID control system configuration previously discussed (Figure 9). A relatively simple LMS filter in conjunction with a PID controller is used to counter cyber-attacks, which improves the performance of the PID controller under cyber-attacks. The LMS algorithm is a widely used adaptive algorithm, particularly in adaptive filtering and neural networks. It is a stochastic gradient descent method that iteratively adjusts filter coefficients to minimize the mean squared error between a desired signal and the output of the filter. This process effectively trains the filter to better match the desired signal characteristics. In Figure 9, the LMS filter block uses the LMS adaptive algorithm. The filter weights or coefficients are required to reduce the error, ε, among the output and the wanted signals. The output signal is the filtered input one, i.e., the approximation of the wanted signal. The error output port gives the outcome of subtracting the output signal from the aimed one. Filtered errors for input and output LMS filters are presented in Figures 10 and 11, accordingly. The input control signal is shown in Figure 12. The trajectory with LMS filtering is shown in Figure 13. In this case, the trend line with the determination coefficient $R^2 = 0.95$ is the approximation of the trajectory by a best-fit line which has an angle of 40 deg. This trend line is closer to the desired course of MASS, and the chaotic leeway off its course is less significant than with only the PID controller. But even in this case, the MASS continues to remain unpredictable and dangerous for other ships.

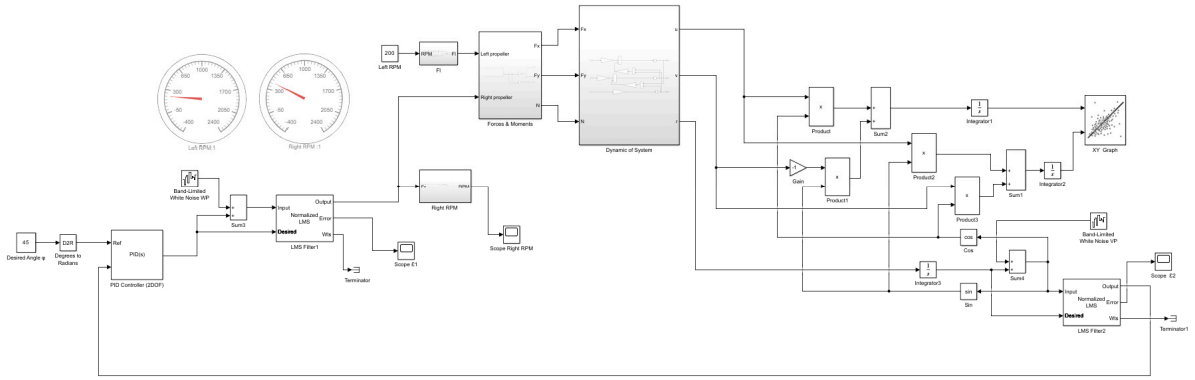


Figure 9. MASS block diagram with PID regulator and LMS filters.

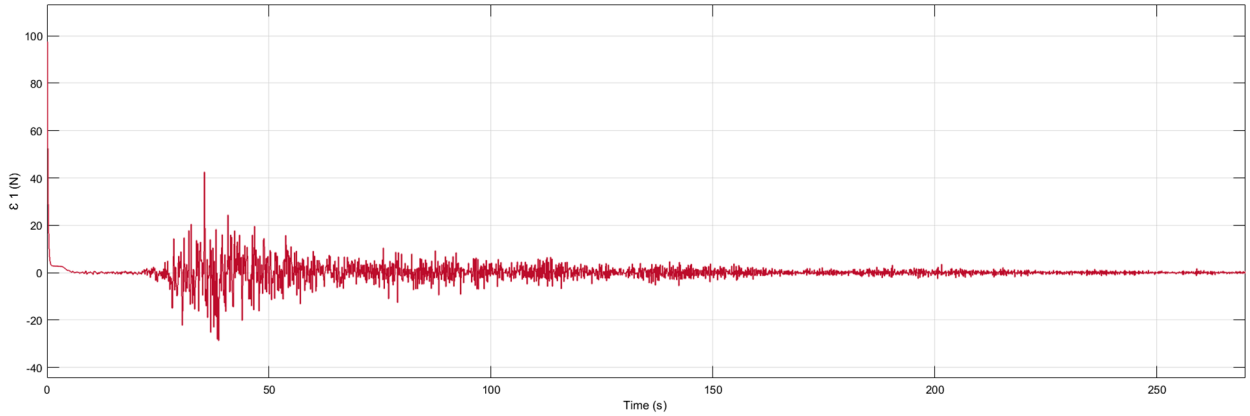


Figure 10. Error for input LMS filter 1.

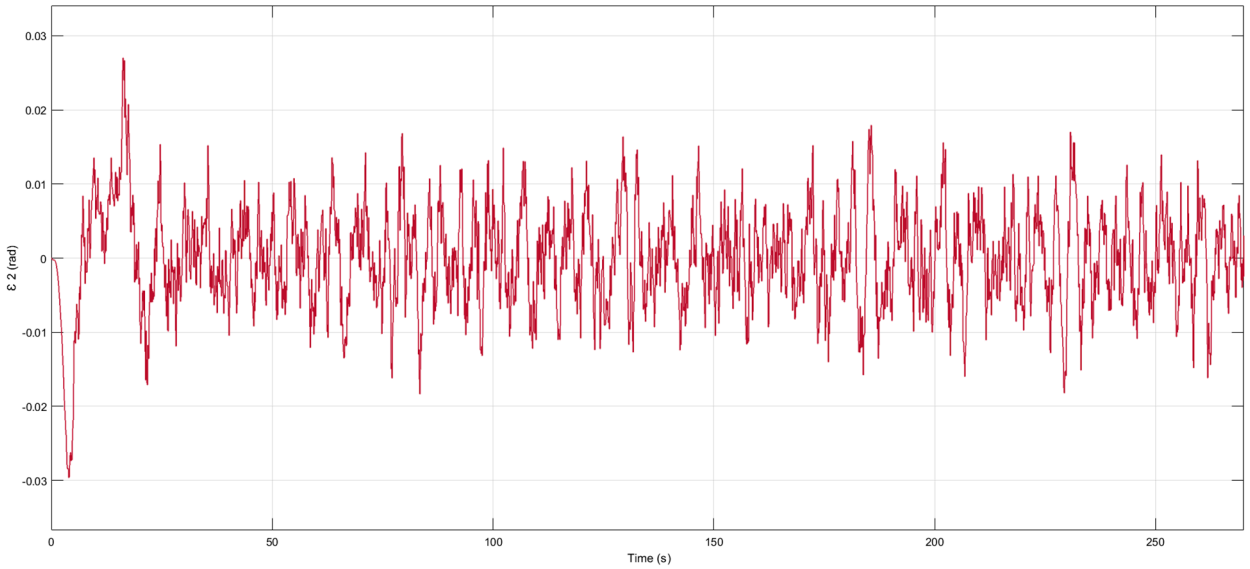


Figure 11. Error for output LMS filter 2.

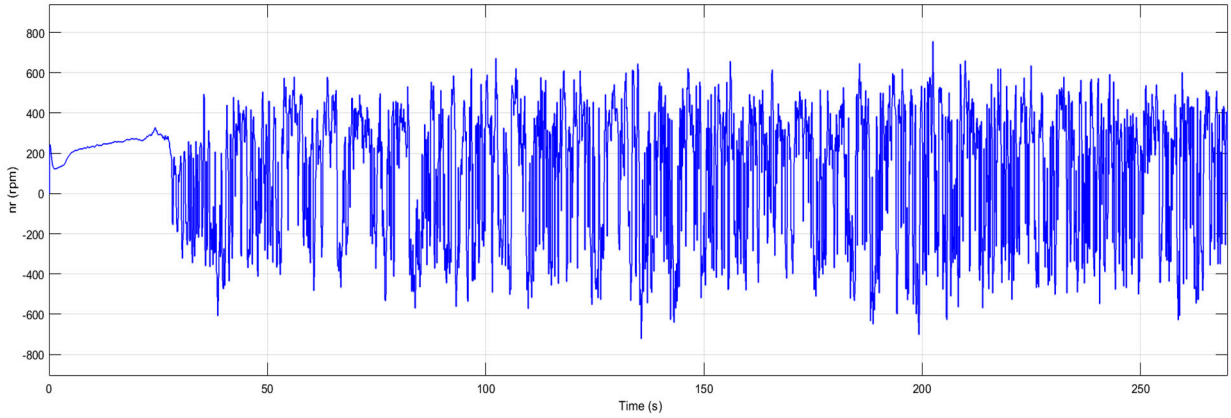


Figure 12. MASS input control signal with LMS filtering.

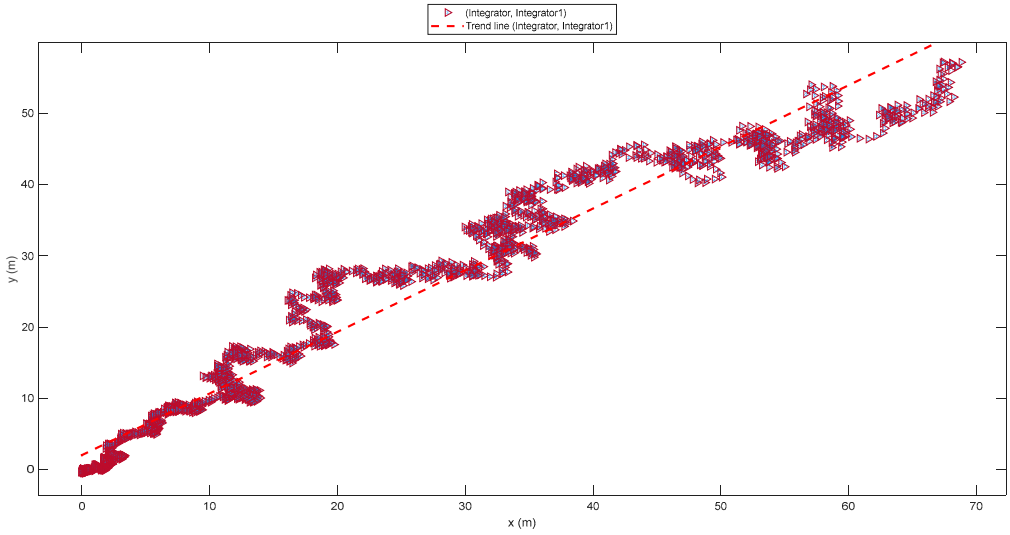


Figure 13. MASS path after using LMS filtering.

3.6. Simulation of Cyberattack Preventing by Using NARMA-L2 Controller

The NARMA-L2 controller is used in the same control system architecture as previously described (see Figure 14). The NARMA-L1 and NARMA-L2 are two approximations to the NARMA model proposed in [54]. Practically speaking, it is discovered that the NARMA-L2 model is easier to set than the NARMA-L1. Here, we grounded our study only on the approximate NARMA-L2 model. For the PID and NARMA controllers under consideration, the logic of detecting a cyber-attack signal is not required. The PID controller has no restrictions on the input and output signals during operation. NARMA has restrictions corresponding to a given input signal, but the output signal is not limited. After tuning, these controllers operate in automatic mode. The internal window, with the associated parameters (Figure 15), can be used to tune the NARMA-L2 regulator. Figure 16 depicts the internal organization of the NARMA-L2 regulator's block, which is depicted in Figure 14. Figure 17 shows the neural network structure from Figure 16.

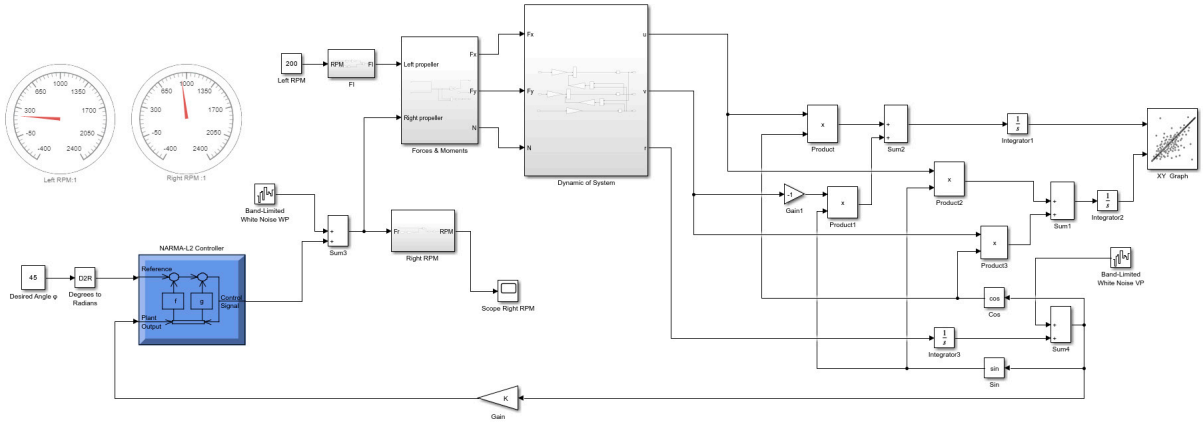


Figure 14. MASS block diagram with NARMA-L2 controller.

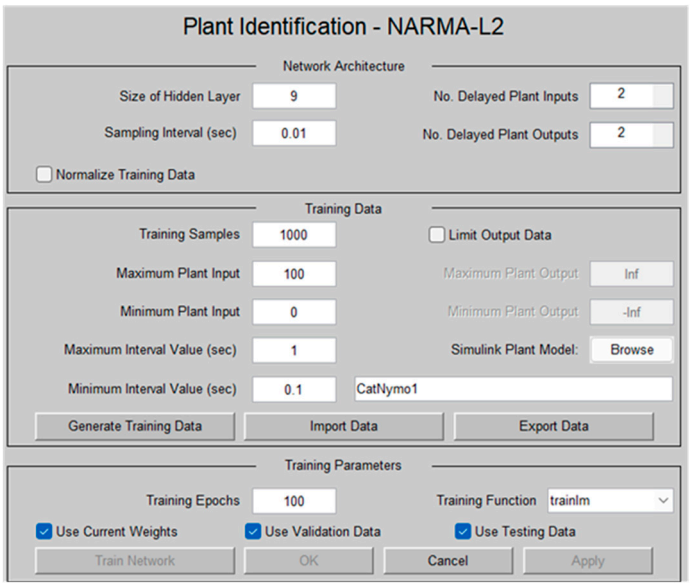


Figure 15. Plant identification—NARMA-L2 window.

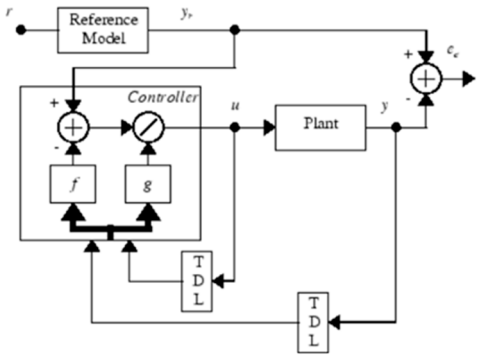


Figure 16. Block diagram of NARMA-L2 controller.

An input control signal is given in Figure 18. The path of the MASS is presented in Figure 19. In this case, the trend line with the determination coefficient $R^2 = 1$ is the approximation of the trajectory by a best-fit line which has an angle of 40 deg. This trend line is closer to the desired course of the MASS, and the MASS makes a smooth and slight leeway off its course. In this case, the MASS remains controllable when exposed to a cyber-attack and is not dangerous to other ships.

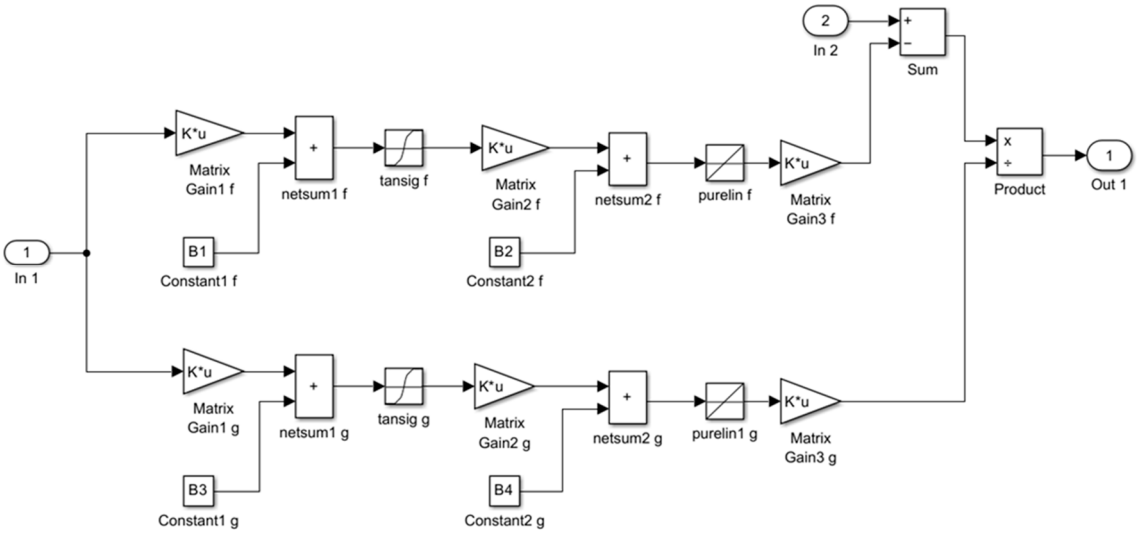


Figure 17. Multilayer neural network.

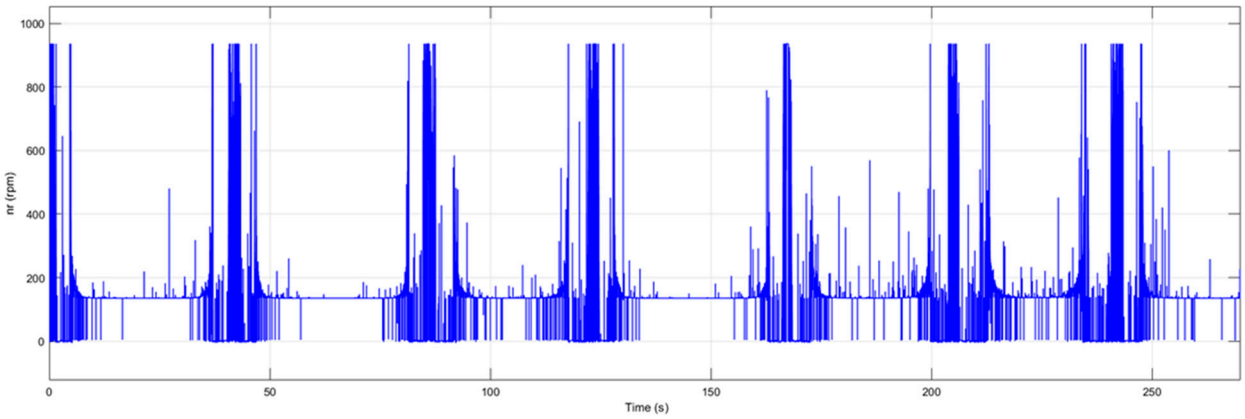


Figure 18. MASS input control signal using NARMA-L2 controller.

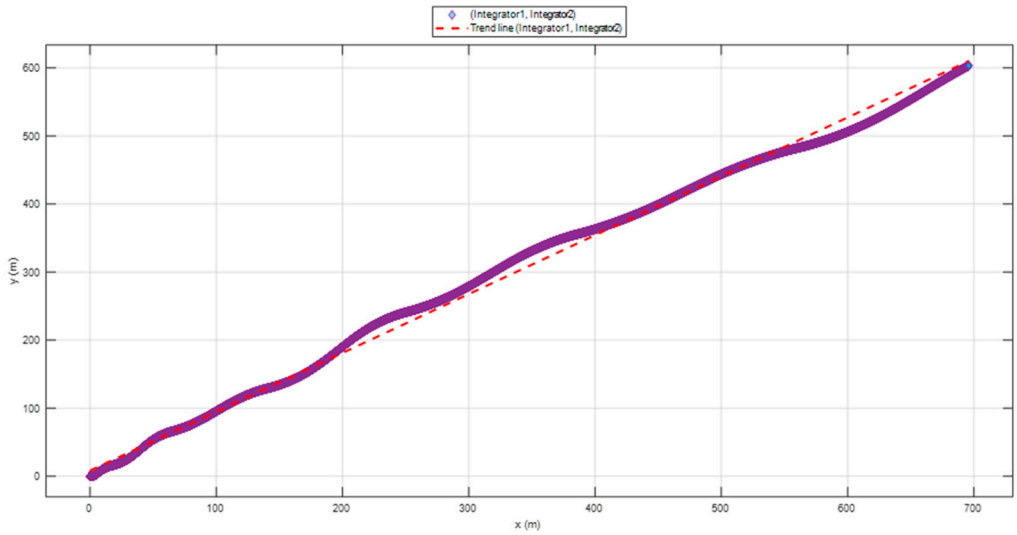


Figure 19. MASS path using NARMA-L2 controller.

4. Discussion

The present study explored the performance differences between conventional PID and NARMA-L2 controllers in the context of cyber-attacks on MASS, specifically applied to

the catamaran-type vessel "Nymo". The results demonstrated a marked contrast between the two control strategies under adversarial conditions, offering valuable insights into both theoretical and applied aspects of autonomous maritime navigation.

From the simulations, it was evident that the traditional PID controller, while effective under nominal conditions, failed to maintain stability and course accuracy under the influence of cyber-attacks simulated as Gaussian white noise disturbances. This aligns with prior findings in control systems literature, where PID controllers are recognized for their limitations in nonlinear or high-disturbance environments. Even with the introduction of LMS filtering to suppress the noise, the PID-controlled MASS still exhibited unpredictability and significant deviation from its desired trajectory, posing potential hazards to surrounding maritime traffic.

In contrast, the NARMA-L2 controller demonstrated robust performance, retaining smooth trajectory control and only minor leeway even under simulated cyber threats. The model's innate capacity to learn and adjust to nonlinear system dynamics and disturbance patterns is responsible for this better performance, which supports earlier studies on brain adaptive control in dynamic contexts.

These findings highlight the critical importance of employing adaptive, learning-based control architectures in the advancement of resilient autonomous waterborne systems. The NARMA-L2 controller's resilience suggests that future implementations of MASS should consider integrating ML/AI methodologies not only for navigation but also for real-time threat mitigation and fault tolerance.

Moreover, the research confirms the practicality of using simulation environments such as Simulink/MATLAB for high-fidelity modeling and control testing of autonomous vessels. This supports earlier work by the authors and others on using model-based approaches to reduce real-world testing costs and risks.

Looking ahead, future research can go in several promising directions:

- Expanding the control strategy to multi-vessel coordination under adversarial conditions.
- Integrating conditions at sea, such as wind, waves, and currents, into the model for more comprehensive realism.
- Applying online learning techniques to NARMA-L2 or alternative artificial neural network architectures to enable real-time adaptation to evolving threats.
- Investigating hybrid control architectures combining neural networks with robust or sliding-mode controllers for improved safety guarantees.

By addressing these directions, researchers can further bridge the gap between theoretical robustness and real-world applicability, ultimately advancing the safe and reliable deployment of autonomous vessels in complex maritime environments.

5. Conclusions

In this paper, a mathematical model for the catamaran "Nymo" maneuvering is presented, and PID and NARMA-L2 controllers for desired heading control are applied, with the objective of maintaining the automatic course of a catamaran under the influence of cyber-attacks.

Simulation results in Simulink are presented. When the obtained results are examined, it is observed that control using conventional controllers, such as PID, shows underperformance when subject to cyber-attacks. In contrast, the NARMA-L2 controller, an artificial neural network-based solution, demonstrates strong resilience and high performance under adversarial conditions.

The resulting simulation model of the "Nymo" catamaran makes it possible to avoid difficult-to-implement experiments in rough sea conditions, offering a safe and cost-effective virtual environment for testing advanced control strategies.

By explicitly modeling the attack vector at the control input level and comparing adaptive versus neural strategies, this study contributes a detailed and practical evaluation of steering-specific cyber resilience, an area not sufficiently addressed in current MASS literature.

However, while the proposed approach shows promise in a simulated setting, there is limited discussion on the real-world feasibility of implementing such neural control systems in operational MASS. The practical challenges of hardware integration, computational overhead, and robustness in diverse environmental and traffic conditions remain open. Additionally, scalability for larger, more complex vessels with different propulsion systems and mission profiles is yet to be evaluated.

Forthcoming research might lead in the following key directions:

- Real-world validation and deployment of the NARMA-L2 control scheme on physical platforms, assessing performance in the presence of real sensor noise, latency, and actuator constraints.
- Scalability studies to examine how the proposed control framework performs with larger vessels or under more complex maneuvering tasks, such as collision avoidance and dynamic route re-planning.
- Integration of ML/AI-based attack detection and mitigation systems (e.g., anomaly detection using unsupervised learning, reinforcement learning-based adaptation) to complement control-layer robustness with situational awareness and proactive defense.
- Hybrid control architectures that combine neural networks with classical or robust control elements to ensure interpretability, safety, and compliance with maritime regulations.
- Modular simulation environments that allow for plug-and-play experimentation with various attack models, control algorithms, and vessel types, thereby promoting repeatable and extensible research.

Pursuing these directions will significantly enhance the real-world applicability and scientific impact of autonomous maritime systems in the face of growing cybersecurity threats.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
(S-)AIS	(Satellite-)Automatic Identification System
AUV	Autonomous Underwater Vehicle
BIMCO	Baltic and International Maritime Council
COLREG	Convention on the International Regulations for Preventing Collisions at Sea
CPS	Cyber Physical System

DDoS	Distributed Denial of Service
EGNOS	European Geostationary Navigation Overlay System
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
HAZID	Hazard Identification
IMO	International Maritime Organization
ISPS	International Ship and Port Facility Security code
IT	Information Technology
LMS	Least Mean Squares
LUSV	Large Unmanned Surface Vehicle
MASS	Maritime Autonomous Surface Ship
ML	Machine Learning
NARMA-L1	Nonlinear Autoregressive Moving Average with Exogenous Input-L1 neural network controller
NARMA-L2	Nonlinear Autoregressive Moving Average with Exogenous Input-L2 neural network controller
NED	North–East–Down
OT	Operation Technology
PID	Proportional–Integral–Derivative
SCADA	Supervisory Control and Data Acquisition
STPA	System Theoretic Process Analysis
STRIDE	Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege
XLUUV	Extra Large Unmanned Undersea Vehicle

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